Created By:

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BT503 Final + Mid Current Solved

Mid Term Mcqs

MCQS

1. Natural estrogens deactivated by .....**Glucuronidation**
2. Insect virus that causes over expression of genes in molecular biology .....**baculo virus**
3. . Acute toxins..... **All**
4. By using recombinant technologies scientists are now producing..... **Insulin**

5.are combustible materials, containing within themselves all oxygen needful for their combustion, which burn but do not explode, and function by producing gas which produces an explosion…..**Propellants or low explosives**

6.are types of waste stabilization pond used for biological treatment of industrial and domestic wastewater..... **Facultative lagoon**

7. PAH-containing fractions, by animal test systems..... evidence have been obtained that PAH containing..... **(skin painting, lung implantation)**
8. Pentane is utilized as a ……in the preparation of phenol..... **foaming**
9. Polystyrene, as a propellant for aerosol sprays, as a filling of low-temperature **thermometers**, and as reference compound in gas chromatography.
10. Herbicide resistant plants ..... **pests**
11. Average temperature of ocean…..... **1–3°C**
12. Which of ....does not use to make vessel of biofilters.... **(may be formic sheet)**
13. Which of the following is not fate of water .....**(may be generation of aerosol)**
14. Which of the following does not classify as a substitution material..... (other than those options) **materials is a major potential avenue for the environmental biotechnology**
15. Which of the following statement about desulphurization of oil and coal is false.....**(See the ppt page no 390 to 400)**
16. Metals are released into the environment by all of the following except **(Earthquake)**
17. When facultative pond receives raw sewage they are called.....ponds. **(primary pond)**
18. Which of the following is not linked to biomagnifications
19. Why frogs are helpful as bioindicators..... **Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats**
20. For land base application soil type best suited to biotechnological intervention are..... **(Sand and gravel)**
21. Attack by insect not only cause damage to the plant but also provide route..... **for bacterial or fungal infection**
22. Insertion of foreign DNA in the vector resulting in the loss of the future. **(marker gene)**
23. Methogenic contortion of over 100 bacterial clone with the capability to digest ... **(terephthalate)**
24. Preliminary treatment involves the removal of coarse…… **solids .....** only
25. Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. **If an indefinite population growth**

26. Septic tanks are also a form of ,,,,,,,,,**primary** ,,,,,,,,,treatment
27. Mere a gene is one of a ……..**cluster**.....of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form
28. Facultative ponds which receive raw sewage is also known as..... **primary ponds**
29. Facultative ponds which receive raw sewage is also known as primary ponds .....**Retention by the plants**
30. In the Inhofe tank, settling occurs in the… .....**upper compartment**
31. Second strategy is usually .....**plasmid**
32. Fundamental requirements are.....**all of given above**
33. Attack by insects is cause……**micrococcus lentos**
34. Na+/H+ antiport, ….**the transport of sodium and hydrogen ions** …in opposite directions across a membrane
35. Frog are bio indicator by ..... **Change in fresh and terrestrial**
36. For higher efficiencies of phosphorus removal, effluent polishing methods can adopted:
    Addition of coagulants
    (Metallic ions):
    Phosphorus precipitation
    Effluent filtration: removal of the phosphorus present in the suspended solids in the effluent
    Combination of the addition of coagulants and filtration
37. Wastewater is applied to the soil, supplying water and nutrients necessary for **plant growth.**
38. Bio filter consists of a relatively large vessel or container, typically made of cast concrete, metal or **durable plastic**
39. The metals released in air by the processes of weathering, erosion and volcanic activity
40. The nature of the soil or water which harbors the pollution can have a major effect on the actual expressed end-result.
41. Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, instead relying on tides and currents as a transport mechanism. They feed upon phytoplanktons, bacterioplanktons, or detritus (i.e. marine snow).
42. Phytoplanktons, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop.
43. Mere a gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form.
44. The insect virus, Baculovirus, has been shown to be the method of choice for the over expression of genes in many applications of molecular biology.
45. A strain of Pseudomonas putrid known to degrade PCBs
46. Facultative ponds which receive raw sewage is also known as primary ponds
47. Bio substitution are: **Biofuels, Polymers, Biodegradable**
48. septic tank can be a single-chamber tank or a two-compartment tank **(called an Imhoff tanks).**
49. In the Imhoff tank, settling occurs in the upper compartment **(settling compartment).**
50. In the soil, a pollutant has basically four possible destinations:

- **retention in the soil matrix,**
- **retention by the plants,**
- **appearance in the underground water,**
- **collection by underdrains**

51. Facultative ponds which receive raw sewage is also known as **primary ponds**

52. A strain of Pseudomonas putida known to **degrade PCBs**

53. Which of does not use to make vessel of biofilters.....**(may be formic sheet)**

54. Which of the following is not fate of water.....**(may be generation of aerosol)**

55. Which of the following does not classify as a substitution material..... **(other than those options) materials is a major potential avenue for the environmental biotechnology**

56. Which of the following statement about desulphurization of oil and coal is false..... **(See the ppt page no 390 to 400)**

57. Metals are released into the environment by all of the following except..... **(Earthquake)**

58. When facultative pond receive raw sewage they are called ponds.....**(Primary pond)**

59. Which of the following is not linked to biomagnifications **…(other than those) • long-lived, • mobile, • soluble in fats, • biologically active.**

60. Why frogs are helpful as bioindicators … **Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats**

61. For land base application soil type best suited to biotechnological intervention are **(Sand and gravel)**

62. **The septic tank can be a single-chamber tank or a two-compartment tank (called an Inhofe tank)**.

63. In the Inhofe tank, settling occurs in the upper compartment **(settling compartment)**.

64. Attack by insect not only cause damage to the plant but also provide route … **for bacterial or fungal infection**

65. Insertion of foreign DNA in the vector resulting in the loss of the future..... **(marker gene)**

66. Methogenic contortion of over 100 bacterial clone with the capability to digest …**(terephthalate)**

Preliminary treatment involves the removal of coarse .....**solids only**

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Short Question

What is the bio indicator?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment. The expression 'Bioindicator' is used as an aggregate term referring to all sources of biotic and abiotic reactions to ecological changes.They are used for assessing environmental health and biogeographic changes taking place in the environment e.g.,

a plankton responding rapidly to changes taking place in the surrounding

Importance of an aerobic zone?

It is essential to have anaerobic and aerobic zones in the treatment line for the biological removal of phosphorus. The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Bio filter? What are the bio filter give example?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

Examples of biofiltration include: Bioswales, biostrips, biobags, bioscrubbers, Vermifilters and trickling **filters**.

Constructed wetlands and natural wetlands. Slow sand **filters**.

Glyphosate?

'Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5-enolpyruvylshikimate-3-phosphate synthase. The gene coding for this enzyme has been identified, isolated and inserted into a number of plants

e.g., Petunias. The gene was expressed behind a CaMV promoter and introduced using A. tumefaciens.

How we improve crop quality?

Consider these 9 tips, techniques, and methods regarding how to increase your corn crop yield

1. Plant Early, Plant Effectively. ...
2. Practice Seasonal Soil Rotation. ...
3. Know The Yield Potential. ...
4. Always Scout Your Fields. ...
5. Ensure Proper Water Drainage. ...
6. Utilize Fertilizers. ...
7. Test Your Soil. ...
8. Weed Early and Often.

Two advantage of conventional system of sludge?

Advantages of Conventional Dispersed Plug Flow Process 1. Allows smaller volume than CSTR 2. Flexible operation, zone aeration, step feeding options, accommodates anoxic and aerobic processes with single biomass for biological nutrient removal 3. Less aeration than CSTR 4.

How marine plant indicate the health of environment give example?

Marine plants provide valuable information to predict the status of oceanic environment, as they are immobile and rapidly obtain equilibrium with their natural surrounding.

Changes in the diversity of species of phytoplankton, including Euglena clastica, Phacus tortus, and Trachelon anas, indicate the pollution of marine ecosystems

What are phase of bio filters?

The first methods to be developed. Consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic, which holds a filter medium of organic material such as peat, heather, bark chips and the like.The gas to be treated is forced, or drawn, through the filter.

Three phase of bio filter?

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This biofilter is a three-phase reactor composed of: Solid phase: consists of a support medium and biofilms, Liquid phase: consists of the liquid in permanent flow through the porous medium, Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Transformation of plant resistance against herbicide?

The transfer of **herbicide resistance** genes from various sources into crop **plants** has been performed using several techniques. These transgenic products rely on both target site and metabolic detoxification **resistance** mechanisms.

Acidophile?

Acidophiles or **acidophilic** organisms are those that thrive under highly acidic conditions (usually at pH 2.0 or below). These organisms can be found in different branches of the tree of life, including Archaea, Bacteria, and Eukarya.

Phytoplankton?

Phytoplankton are microscopic marine algae. **Phytoplankton** is the base of several aquatic food webs. In a balanced ecosystem, they provide food for a wide range of sea creatures including whales, shrimp, snails, and jellyfish.

Function of bio trickling filter over bio filter?

A **biotrickling filter** is a combination of a **biofilter** and a bioscrubber. The bacteria responsible for decomposition are immobilised **on** a carrier or **filter** material. .....This water also carries away excess sludge/biofilm, as well as decomposition products which may hinder the biomass.

Role of phytoplankton for the transfer of pollutant among the different trophic level?

What is the role of phytoplankton in the transfer of pollutants among different trophic level?

Phytoplankton are the primary producers **in** aquatic ecosystems and play a key **role in** the **transport** of organic contaminants through the **food chain** to higher **trophic levels .....** The bioaccumulation at lower **trophic levels** forms the first step **in** the **transfer** of chemicals through the **food chain**.

Advantage of bio indicator?

The most important reasons for using **bioindicators** are: the direct determination of **biological** effects, the

determination of synergetic and antagonistic effects of multiple pollutants on an organism, the early recognition of pollutant damage to plants as well as toxic dangers to humans and.

Submerged bio filter?

Besides being a support medium for biomass growth, the granular material acts also as a filter medium. Periodic washings are necessary to eliminate the excess biomass accumulated.

Biotechnology and its all applications?

Biotechnology has **applications** in four major industrial areas, including health care (medical), crop production and agriculture, non-food (industrial) **uses** of crops and other products (e.g. biodegradable plastics, vegetable oil, biofuels, and environmental **uses**).

Pollutant?

A pollutant is a substance or energy introduced into the environment that has undesired effects, or adversely affects the usefulness of a resource.

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Bio reactor?

a bioreactor is a vessel in which a chemical process is carried out which involves organisms or biochemically active substances derived from such organisms.

Glysophates?

Glyphosate is a broad-spectrum systemic herbicide and crop desiccant. It is an organophosphorus compound, specifically a phosphonate, which acts by inhibiting the plant enzyme 5-enolpyruvylshikimate-3-phosphate synthase. It is used to kill weeds, especially annual broadleaf weeds and grasses that compete with crops.

List two advantage of conventional activated sludge system?

High biomass concentration in the reactor due to the recirculation of the settled bacteria from the bottom of the secondary sedimentation tank. Biomass remains in the system longer than the liquid, resulting in high BOD removal efficiency.

Environmental biotechnology?

Environmental biotechnology is biotechnology that is applied to and used to study the natural environment. Environmental biotechnology could also imply that one try to harness biological process for commercial uses and exploitation.

How can be prepared plant as recombinant?

In simpler terms, **recombinant DNA** is formed when the **DNA** from one organism is incorporated into the **DNA** of another organism of the same or different species.....Using enzymes like ligase, the cut **DNA** fragment is attached to the vector. This is then introduced into the host cell, which takes it up as part of its own **DNA**.

How plants are modified as bio indicator?

Plant indicators The presence or absence of certain **plant** or other vegetative life in an ecosystem can provide important clues about the health of the environment: environmental preservation. There are several types of **plant** biomonitors, including mosses, lichens, tree bark, bark pockets, tree rings, and leaves.

What are thermophile? Where are they found?

A thermophile is an organism—a type of extremophile—that thrives at relatively high temperatures, between 41 and

122 °C. Many thermopiles are archaea. Thermophilic eubacteria are suggested to have been among the earliest bacteria.**Thermopiles** are **found** in various exothermally heated regions of the Earth, such as hot springs like those in Yellowstone National Park (see image) and deep sea hydrothermal vents, as well as decaying plant matter, such as peat bogs and compost.

Three phase submerged aerated?

This biofilter is a three-phase reactor composed of: Solid phase: consists of a support medium and biofilms, Liquid phase: consists of the liquid in permanent flow through the porous medium, Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

Animal as bio indicator and give example?

A **bioindicator** is a living organism that **gives** us an idea of the health of an ecosystem.....The numbers of earthworms in the soil can also be used to indicate the health of the soil. **One example** of a **bioindicator** is lichens.

What is the bio indicator?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

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The expression 'Bioindicator' is used as an aggregate term referring to all sources of biotic and abiotic reactions to ecological changes. They are used for assessing environmental health and biogeographic changes taking place in the environment e.g., a plankton responding rapidly to changes taking place in the surrounding

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Long 5 Marks

Three phases of submerged aerated biofilter? Three phases of biofilm?

This biofilter is a three-phase reactor composed of:
Solid phase: consists of a support medium and biofilms,
Liquid phase: consists of the liquid in permanent flow through the porous medium,
Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity

Genetic modification of popular plants how they remove mercury from plants?

Genetic modification of **plants** involves adding a specific stretch of DNA into the plant's genome, giving it new or different characteristics. This could include changing the way the plant grows, or making it resistant to a particular disease.

There are two types of **mercury removal** materials: non-regenerative sorbents and regenerative adsorbents. With non-regenerative types of **mercury** sorbents, the process fluid flows through the sorbent bed for a number of years, after which the sorbent is replaced.

Difference between transformation, transduction and conjugation?

In **transformation**, a bacterium takes up a piece of DNA floating in its environment. In **transduction**, DNA **is** accidentally moved from one bacterium to another by a virus. In **conjugation**,
DNA **is** transferred **between** bacteria through a tube **between** cells.

Function of filter medium in bio filter and of compound used in?

The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity

Recombinant microorganism particularly bacteria and yeast?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.
Many examples… A strain of Escherichia coli into which was engineered some 15 genes originating from Pseudomonas.
These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.

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Current Final Term

BT 503 Final Current Solved – Mostly Question Solved from Internet

Question and Answers

Feature of phenol that make it pollutant?

Phenols and some detergents may also mimic the activity of oestrogen.

Name enzyme equipment and solution used in genetic engineering?

the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell. It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Biological process of phospherial removal activated sludge treatment?

The **process** of **removing phosphorus** by accumulating it with biomass is referred to as enhanced **biological phosphorous removal** (EBPR) **process**. The **phosphorous removal** efficiency for **biological** systems depends on the **phosphorus** content of the **sludge** removed and the efficiency of the solids separation **process**. **(Internet)**

When the biological excess sludge is wasted from the system, phosphorus is removed… For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: • addition of coagulants (metallic ions): phosphorus precipitation • effluent filtration: removal of the phosphorus present in the suspended solids in the effluent • combination of the addition of coagulants and filtration. **(PPTS)**

Photodegradation and name its forms?

Photodegradation involves the **process** of oxidation (loss of electrons, increase in oxidation state) and hydrolysis (split of chemical bonds by adding water).....**Photodegradation** does not talk about the ageing or degradation of materials with the use of heat or infrared light.

What are living conditions extremophile give example?

Three **examples** of **extremophiles** are Picrophilus torridus (a thermoacidophile adapted to hot, acidic **conditions**), Antarctic krill (a psychrophile), **and** the Pompeii worm (a thermophile).

What are application of biosensor in detection of biocides?

Biosensors have been extensively applied in clinical, food and environmental areas due to the advantages of fast **detection** speed, high selectivity and sensitivity (Malhotra and Chaubey, 2003). In this study, the contribution of bioluminescence and luminometer makes a **biosensor** for **detection** of water pollution.

Explain genomic libraries?

A **genomic library** is a collection of the total **genomic** DNA from a single organism. The DNA is stored in a population of identical vectors, each containing a different insert of DNA. .....The fragments are then inserted into the vector using DNA ligase.

Biofilter?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants. Common uses include processing waste water, capturing harmful chemicals or silt from surface runoff, and microbiotic oxidation of contaminants in air.

Biodiesel?

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat, soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

Plant associated with fungi?

Mycorrhizae: Associations of fungi with roots of vascular plants

mycorrhizae, mycorrhiza or mycorrhizas) is a symbiotic association between a **fungus** and a **plant**. The term mycorrhiza refers to the role of the **fungus** in the **plant's** rhizosphere, its root system. Mycorrhizae play important roles in **plant** nutrition, soil biology and soil chemistry.

Water system?

water system. a river with all its tributaries. a **system**, including reservoirs and mains, for storing and supplying **water** for use by a community.

How plant can use be reduced pollution?

Plants improve air quality through several mechanisms: they absorb carbon dioxide and release oxygen through photosynthesis, they increase humidity by transpiring water vapor through microscopic leaf pores, and they **can** passively absorb **pollutants** on the external surfaces of leaves and on the **plant** root-soil system.

How animal work as bioindicator two example?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

Example of a **bioindicator** is lichens. These plants, which live on surfaces such as trees or rocks or soil, are very sensitive to toxins in the air.
Greasewood indicates saline soil

Define phytoremediation?

Phytoremediation /ˌfaɪtəʊrɪˌmiːdɪˈeɪʃən/ refers to the technologies that use living plants to clean up soil, air, and water contaminated with hazardous contaminants.

What are biodegradation?

Biodegradation is the breakdown of organic matter by microorganisms, such as bacteria and fungi.

Bacteriophage?

In bacteria, the most likely candidates for genetic transfer are plasmids and bacteriophages.

Biosensor advantage?

Advantages of tissues as **biosensors** include the following: easier to immobilize compared to cells and organelles. the higher activity and stability from maintain enzymes in natural environment.

Sources dioxin?

- **Combustion** sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning.
- Metal smelting.
- Refining and process sources.
- Chemical manufacturing sources.
- Natural sources

Method to get quality crop?

The average **quality** can be increased by planting seeds into fertilized soil. Increasing Farming Skill also increases the chance to harvest a **quality crop**. If a Seed Maker turns a higher-**quality crop** into seeds, the seeds are no more likely to produce higher-**quality crops** when grown and harvested.

PCB TNT and subtreated biofilter?

??

Phase sludge water in sec wastewater treatment?

Sludge is a semi-solid slurry that can be produced from a range of industrial processes, from **water treatment**, **wastewater treatment** or on-site sanitation systems. ... Industrial **wastewater treatment** plants produce solids that are also referred to as **sludge**.

In which filed biotechnology used?

• manufacturing process, • waste management, • pollution control

Thermophile and its used in PCR? Thermophile and application taq polymerase in PCR?

In PCR: DNA Polymerase of Thermus aquaticus (Taq Polymerase), Pyrococcusfuriosus (Pfu Polymerase). Clean energy: bioethanol production, hydrogen production fuel cells, etc.

Define Extremophiles and name three categories?

Extremophiles are animals that live and thrive under extreme environmental conditions

Psychrophiles can tolerate these circumstances.

Halophiles… survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds. Acidophiles thrive in the conditions of low pH, typically below 5

Extremophile in mature?

??

Trickling filter?

A **trickling filter** is a type of wastewater treatment system. It consists of a fixed bed of rocks, coke, gravel, slag, polyurethane foam, sphagnum peat moss, ceramic, or plastic media over which sewage or other wastewater flows downward and causes a layer of microbial slime (biofilm) to grow, covering the bed of media.

Advantage and disadvantage of incineration?

Advantages: • Drastic volume reduction, • Sterilisation

Disadvantages: • High costs, • Ash disposal, • Atmospheric pollution.

Effect of dioxin on human?

dioxins may result in skin lesions, such as chloracne and patchy darkening of the skin, and altered liver function. Long-term exposure is linked to impairment of the immune system, the developing nervous system, the endocrine system and reproductive functions

Long 5 Marks

Explain environmental factor bioremediation?

bioremediation is the use of living organisms, primarily microorganisms, to degrade the **environmental** contaminants into less toxic forms. It uses naturally occurring bacteria and fungi or plants to degrade or detoxify substances hazardous to human health and/or the **environment**.

Detail note on BCD and use of biosensor to detect these?

??

Biodiesel advantage?

Biodiesel is a domestically produced, clean-burning, renewable substitute for petroleum diesel. Using biodiesel as a vehicle fuel increases **energy** security, improves air quality and the environment, and provides safety benefits.

Solid waste step?

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

Genetic modification in popular plant used to removal mercury? Describe how?

Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere.

mer A gene gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one

coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

Initially the gene was transferred to Arabidopsis thaliana. The gene was transferred by gene guns to poplar tree (Liriodendron tulipifera) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatalise it at 10 times the rate observed in untransformed yellow poplar plantlets.

Factor affect bioremediation?

The factors that directly impact on bioremediation are energy sources (electron donors), electron acceptors, nutrients, pH, **temperature**, and inhibitory substrates or metabolites. One of the primary distinctions between surface soils, vadose zone soils and groundwater sediments is the content of organic material.

Properties of soil which has effect on pollution?

The application of waste or **polluted** water into the **soil** alters its physical and chemical, thereby affecting the growth of agricultural crops and other living organisms. Even the extensive agricultural practices such as fertilizer and pesticide application deteriorate the **soil quality**.

Detail note on environmental biotechnology?

Environmental biotechnology. **Environmental biotechnology** is the branch of **biotechnology** that addresses **environmental** problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

Explain recombinant microorganism especially yeast and microorganisms?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples… A strain of Escherichia coli into which was engineered some 15 genes originating from Pseudomonas. These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.

Long 10 Marks

Briefly describe category of solid waste?

Different categories of solid waste include:

- Non-combustibles: Metal, tin cans, bottles, stones, etc.
- Ashes/dust: Residue from fires used for cooking.
- Bulky waste: Tree branches, tyres, etc.
- Dead animals: Carcasses of domestic animals and livestock
- Hazardous waste: Oil, battery acid, medical waste.
- Construction waste: Roofing, rubble, broken concrete, etc.

Genomic Library?

Genomic libraries Collection of total Genomic DNA from an organisms. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell. These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain non-coding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

Sludge treatment procedure?

Sewage **sludge treatment** describes the **processes** used to manage and dispose of sewage **sludge** produced during sewage **treatment**. **Sludge** is mostly water with lesser amounts of solid material removed from liquid sewage. ...

Reducing **sludge** volume may increase the concentration of some of these toxic chemicals in the

sludge.

Phytovolatilization?

Phytovolatilization is a process, in which plants take up contaminants from soil and release them as volatile form into the atmosphere through transpiration. The process occurs as growing plants absorb water and organic contaminants.

What is cloning factor? Why important in genetic engineering?

Cloning allows for the creation of multiple copies of **genes**, expression of **genes**, and study of specific **genes**. ...

Plasmids have been highly **engineered** as vectors for molecular **cloning** and for the subsequent large-scale production of **important** molecules, such as insulin.

Southern blotting and western blotting?

In Southern blotting, DNA fragments are resolved by electrophoresis on a gel. These resolved fragments are then shifted to a membrane (nylon or nitrocellulose).

Western blotting: Proteins (e.g., antibodies antigens interaction)

Advantages and disadvantages of incineration (2)

Incineration:

Advantages:

•Drastic volume reduction,
•Sterilization.

Disadvantages:

•High costs,
•Ash disposal,
•Atmospheric pollution.

Manipulation of bacteria without genetic engineering (3)

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. The 'training' may be required either to improve the bacterium's tolerance to the pollutant or to increase the capabilities of pathways already existing in the bacterium to include the ability to degrade the pollutant, or a combination of both.

3 phases present in submerged bio filter (3)

Typically, a biofilter has two or three phases, depending on the feeding strategy (percolating or submerged biofilter): a solid phase (media); a liquid phase (water); a gaseous phase (air).

Co-metabolism

Co-metabolism is defined as the simultaneous degradation of two compounds, in which the degradation of the second compound (the secondary substrate) depends on the presence of the first compound (the primary substrate).

Three forces are in the global environment market

A number of forces over which it has little or no control affect a company's marketing activities. Taken together, they make up its external marketing environment, which includes regulatory and political activity, economic conditions, competitive forces, changes in technology, and social and cultural influences.

Phytodegradation

Phytodegradation which is also known as phyto-transformation is the breakdown of contaminants taken up by plants through metabolic processes within the plant, or the breakdown of contaminants surrounding the plant through the effect of enzymes produced by the plants.

Phytoremediation

The treatments of pollutants or waste (as in contaminated soil or groundwater) by the use of green plants that remove, degrade, or stabilize the undesirable substances (such as toxic metals).

Three sources of dioxin production

According to the most recent US EPA data, the major sources of dioxins are broadly in the following types: Combustion sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning. Metal smelting. Refining and process sources.

Conjugation transduction plasmid difference

Conjugation is the transfer of circular DNA called plasmids through cell to cell contact. Transformation is the uptake of 'free' DNA from the environment. Transduction is the transfer of DNA by bacteria-specific viruses called bacteriophage.

Biodiesel

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat (tallow), soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.